SCIENCE & INNOVATION FOR SUSTAINABILITY

PROJECT RESEARCH & INVESTIGATION REPORT

KRISHI-DRISHTI (Krishi-Drishti)

An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture

Project Category:	Senior School Science & Innovation Project (Class 9 / Age: 14–15 Years)
Student 1 (Lead Researcher):	[Full Name of Lead Student / Researcher]
Student 2 (Co-Worker):	[Full Name of Co-Worker / Team Member]
Class & Section:	Class 9, Section A
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Guide Teacher / Mentor:	[Name of Guide Teacher, TGT Science]
Project Investigation Period:	July 2026 – November 2026

"We started this project after seeing viral news and farmer videos showing identical crop disasters across multiple states—from tomatoes dumped in Nashik and Kolar to potato fields rotting in Bihar and UP. We wanted to build an affordable, tough, and 100% offline screening wand under Rs 1,500 that any small farmer can hold in their hand right in the field before spraying costly chemicals."

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4. CERTIFICATE & STUDENT DECLARATION

Student Declaration: We hereby declare that the project titled *'KRISHI-DRISHTI: An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture'* is our own original work. We assembled the circuit, tested the code, collected field leaves, and analyzed the errors under the guidance of our teacher. All facts, test results, and failure logs reported here reflect actual physical observations.

Signature of Student 1 (Lead Researcher): _____ **Date:** ____ / ____ / 2026
Signature of Student 2 (Co-Worker): _____ **Date:** ____ / ____ / 2026

Guide Teacher Certificate: This is to certify that [Student 1] and [Student 2], students of Class 9, have carried out this investigation under my supervision. To the best of my knowledge, this report represents their own study, physical prototyping, and field observations.

Signature of Guide Teacher: _____ **Institution Stamp:** [SEAL]

5. PROJECT ABSTRACT

Last year, agricultural news and social media showed devastating crop failures across major tomato and potato belts. In tomato hubs like Kolar and Nashik, early blight and fruit rot wiped out harvests, forcing farmers to dump damaged crates on roads. Just months later, winter fog hit northern India, and potato farmers across Samastipur, Vaishali, and Agra saw their fields collapse into black, rotting slime from Late Blight within 72 hours. Seeing the exact same tragedy happen across places A, B, C, and D, I investigated why farmers could not catch the infection early. In online farmer groups, growers desperately asked what chemicals to spray, but by then spraying was useless. Existing mobile apps fail because 4G cellular signals drop to zero out in open fields, and uncalibrated phone cameras take blurry photos blinded by sun glare.

To solve this, I designed KRISHI-DRISHTI, an ultra-low-cost (under Rs 1,500) offline screening kit. It pairs an ESP32 controller wand (with an OLED screen and a DHT11 canopy humidity sensor) with an ESP32-CAM unit fitted with an 8 cm black optical hood that standardizes focal distance and blocks harsh sunlight. The camera creates an offline Wi-Fi hotspot, sending images to an old Android phone in the pocket that runs a quantized MobileNetV3 AI model in just 4.9 ms without internet.

On 60 clean botanical test leaves, the system achieved 88.33% exact diagnosis (53/60) and 90.0% pathogen-family correlation. On 93 messy outdoor PlantDoc photos, raw accuracy was 33.33%, but dual-stage Shannon entropy gating safely rejected 37 confusing images as UNCERTAIN, raising accepted accuracy to 55.36%. Physical testing revealed real limitations: an unseen tea leaf reached 87% false confidence, and night flashlight shadows mimicked blight. KRISHI-DRISHTI provides an honest, accessible preliminary screening tool that aids farmers in early organic intervention.

Keywords: *Offline Edge-AI, ESP32 Wand, Late Blight, Tomato, Potato, Optical Shroud, Smallholder Farmers.*

6. INTRODUCTION & BACKGROUND

More than seven out of ten rural families in northern India depend directly on agriculture for their livelihood. During the rabi season, potatoes (*Solanum tuberosum*) and tomatoes (*Solanum lycopersicum*) are staple cash crops grown extensively across the fertile Gangetic plains. These crops provide the main seasonal income for smallholder families between November and March.

However, winter also brings dense fog, dripping cold nights, and prolonged overcast skies. Under these exact microclimate conditions, a devastating disease called Late Blight spreads with terrifying speed. It is caused by *Phytophthora infestans*, an oomycete water-mold. If a farmer does not catch the first small lesions on the lower leaves, an entire field can be completely destroyed within 48 to 72 hours. Similarly, Early Blight (*Alternaria solani*) attacks foliage, producing brown concentric rings that weaken plants and ruin tomato and potato harvests.

7. WHY I CHOSE THIS PROBLEM (TRACKING REAL CROP LOSSES ACROSS STATES)

This project started not from a textbook, but from following real agricultural news broadcasts and farmer community channels on social media (including YouTube agricultural channels like Krishi Jagran and farmer WhatsApp groups). Last year, I watched heartbreaking videos of tomato farmers in Kolar (Karnataka) and Nashik (Maharashtra) dumping whole tractor loads of tomatoes onto the roads because severe early blight and fruit rot had ruined their produce.

Just a few weeks later, during winter, identical reports started flooding in from potato belts across northern India: first in Samastipur and Vaishali in Bihar, then Agra in Uttar Pradesh, and later parts of Punjab and West Bengal. Farmers posted videos walking through their fields in shock—healthy green plants had turned into black, rotting slime in less than 72 hours! Seeing this exact same tragedy happen at place after place—Place A, Place B, Place C, and Place D—got me investigating.

In the video comment sections, hundreds of farmers desperately asked what chemicals to spray. Most sprayed expensive, unverified fungicides blindly after the plants were already dead, wasting thousands of rupees. When I checked existing mobile apps, I saw that they require 4G cloud internet—which completely fails when you walk into open fields where mobile network bars vanish. On top of that, unguided phone cameras take blurry photos blinded by sun glare. That was the turning point. I decided to build a practical, ultra-affordable (under Rs 1,500), 100% offline screening wand that a farmer can carry right into the mud to spot diseases early without internet.

8. LOCAL AGRICULTURAL RELEVANCE (RURAL SMALLHOLDER CONTEXT)

In our region, over 85% of farmland is cultivated by marginal farmers owning less than two hectares. These growers face two harsh constraints that commercial technology companies ignore:
1. **Zero Cellular Connectivity in Open Fields:** Walking just 500 meters away from main roads causes 4G/5G mobile signals to drop to zero. Cloud-based agritech apps simply freeze and display connection timeout errors.
2. **High Diagnostic Laboratory Costs:** Sending leaf samples to an agricultural university lab takes 10 to 14 days and costs thousands of rupees. A smallholder cannot wait two weeks when an oomycete infection destroys a crop in three days.

9. PROBLEM IDENTIFICATION IN THE FIELD

Through discussions with 12 local vegetable growers, we identified three clear practical bottlenecks:
• *Visual Confusion:* Early water-rot looks almost identical to sunburn or physical bruising in dim dawn light.
• *Chemical Poisoning:* Out of panic, farmers spray toxic fungicide cocktails at double doses, poisoning soil biology and wasting money.

• *Blurry Phone Cameras:* Without a physical guide, farmers take photos too close or tilted, causing blur, glare, and wild AI errors.

10. RESEARCH QUESTION

"Can a student-built, low-cost handheld wand (costing under Rs 1,500) that pairs an ESP32 microcontroller with a local offline edge-AI model and microclimate sensor provide reliable preliminary foliar health screening for small farmers without needing any internet connection?"

11. AIM AND PRACTICAL OBJECTIVES

Project Aim: To design, assemble, program, and rigorously test an ultra-low-cost, offline-first handheld diagnostic wand and canopy microclimate monitor for early Solanaceae blight screening.

Our Specific Objectives:
1. Build a handheld controller wand using an ESP32 DevKit V1 with pushbuttons, an SSD1306 OLED display, and a DHT11 sensor.
2. Build an optical camera unit using an AI-Thinker ESP32-CAM fitted with an 8 cm shroud to fix working distance and block blinding sunlight.
3. Set up a local Wi-Fi mesh (SoftAP) so the camera, wand, and a pocket smartphone communicate instantly without internet or SIM cards.
4. Quantize a MobileNetV3-Small neural network into Float16 precision (under 2.1 MB) so it runs in under 15 ms on budget phone processors.
5. Program an uncertainty filter using Shannon entropy to reject blurry photos, soil, fingers, and alien leaves instead of guessing blindly.
6. Audit our training dataset cryptographically to ensure zero benchmark leakage and separate true 6-class accuracy from pathogen-family matching.

12. SCIENTIFIC HYPOTHESIS

Null Hypothesis (H0): Using a physical distance shroud and local canopy humidity sensing will not produce any measurable improvement in offline leaf classification accuracy or false-alarm rejection compared to an uncalibrated smartphone camera.

Alternative Hypothesis (H1): Enforcing fixed macro geometry with an optical shroud, filtering out confusing frames with Shannon entropy, and cross-checking foliar spots against canopy relative humidity will significantly improve diagnostic reliability and prevent false alarms under real offline field conditions.

13. BACKGROUND SCIENTIFIC PRINCIPLES

To ensure our project was scientifically sound, our guide teacher helped us study two fundamental biology concepts:

A. The Plant Disease Triangle: In plant pathology, a disease cannot happen just because spores are floating in the air. It requires three factors at the exact same time: a *Susceptible Host* (the potato or tomato plant), a *Virulent Pathogen*, and a *Conducive Environment*. For Late Blight (*Phytophthora infestans*), spores cannot germinate unless canopy relative humidity stays above 85% with temperatures between 18°C and 24.5°C for several continuous hours (Wallin & Hyre models). This is why we wired a DHT11 sensor straight onto our wand: if the air is bone-dry (RH < 60%), a brown spot on a leaf is almost certainly not active Late Blight, preventing unnecessary spraying!

B. Oomycete Water Molds vs True Fungi: Most farmers call everything 'fungus', but Late Blight is actually caused by an Oomycete (a water mold closely related to brown algae). Its cell walls are made of cellulose and beta-glucans, not chitin. Early Blight, on the other hand, is caused by *Alternaria solani*, a true fungus with chitinous cell walls. Common fungicides that target chitin biosynthesis are completely useless against Late Blight. If a farmer sprays a chitin-inhibitor on Late Blight, they are throwing money into the wind.

14. EXISTING APPROACHES AND THEIR LIMITATIONS

Approach	Cost	Internet Needed?	Why It Fails for Small Farmers
Farmer Naked-Eye Guessing	Free	No (Offline)	Very subjective; early water-rot looks identical to physical bruising.
Commercial Smartphone Apps	Rs 500–1,500/yr	Yes (4G/5G Cloud)	App hangs in remote fields; uncalibrated camera blur causes wild errors.
University Lab Testing (PCR)	Rs 15,000–30,000	No (Physical Lab)	Takes 10–14 days. By the time results arrive, the crop is already dead.
KRISHI-DRISHTI (Our Wand)	**Rs 1,480 BOM**	**100% Offline Mesh**	Instant 4.9 ms check, optical shroud fixes focus, checks canopy humidity.

15. OUR PROPOSED SOLUTION

Instead of expecting farmers to buy expensive imported gadgets or carry laptop computers into muddy fields, we built KRISHI-DRISHTI as a lightweight, modular system consisting of three simple physical units:
1. **Handheld Controller Wand:** An ESP32 microcontroller box with two pushbuttons (SCAN and NEXT), an SSD1306 OLED screen, and a DHT11 sensor sticking out of the base to measure microclimate inside the leaf canopy.
2. **Optical Camera Wand:** An AI-Thinker ESP32-CAM fitted with an 8 cm cylindrical black hood (3D-printed or dark PVC pipe) and a diffused white LED. When placed against a leaf, it locks the focus distance at 8 cm and completely blocks harsh sun glare.
3. **Pocket Edge-AI Engine:** An old, inexpensive Android smartphone running in the farmer's pocket. It acts as an offline neural accelerator over a local Wi-Fi hotspot created by the camera, executing inference in just 4.9 milliseconds.

16. COMPLETE SYSTEM ARCHITECTURE AND DATA FLOW

```
[FARMER PRESSES 'SCAN' BUTTON ON WAND]
  |
  v (GPIO 27 Active-LOW Trigger)
[ESP32 CONTROLLER WAND]
  | -- Reads DHT11 (Temp & RH) & Sends HTTP GET /api/capture via Local Wi-Fi
  v
[ESP32-CAM UNIT @ 192.168.4.1]
  | -- Turns on diffuse LED, snaps UXGA 1600x1200 JPEG, saves to MicroSD
  | -- Sets status: 'READY_FOR_INFERENCE' with unique scan_id
  v
[POCKET ANDROID PHONE (OFFLINE PYTHON DAEMON)]
  | -- Fetches raw JPEG over local Wi-Fi (No Internet!)
  | -- Center-crops 224x224 & runs MobileNetV3-Small (Float16 TFLite)
  | -- Computes Shannon Entropy: If H > 1.45 nats, flags 'UNCERTAIN'
  | -- Posts diagnosis + agronomic advice back via HTTP POST /api/diagnosis/set
  v
[ESP32 CONTROLLER WAND]
```

| -- Polls /api/diagnosis/latest & updates SSD1306 OLED
v
[FARMER READS 4-PAGE CAROUSEL ON SCREEN: Disease, Confidence, Weather, & Organic Prescription]

17. HARDWARE COMPONENTS AND EXACT COSTS

Component Name	Key Specification	Source / Market	Cost (Rs)
ESP32 DevKit V1	Dual-core 240MHz, 520KB SRAM, Wi-Fi/BLE	Electronics shop, Patna	Rs 420
AI-Thinker ESP32-CAM	ESP32-S + 4MB PSRAM + OV2640/OV5640	Online hobby store	Rs 540
SSD1306 OLED Display	0.96 inch, 128x64 pixels, I2C interface	Electronics shop, Patna	Rs 190
DHT11 Sensor	Temp (0–50°C), Relative Humidity (20–90%)	School science kit	Rs 95
MicroSD Card (16GB)	Class 10 FAT32 local image storage	Local mobile repair shop	Rs 160
Optical Shroud Tube	8 cm black PLA cone (or 3-inch dark PVC)	DIY fabricated in lab	Rs 45
Pushbuttons, Wires, Cap	2 tact switches, 470uF cap, breadboard wires	School physics lab	Rs 30
Total System BOM Cost	**Complete 2-Wand Screening Kit**	**Affordable to Marginal Farmers**	**Rs 1,480**

18. SOFTWARE AND EDGE-AI ARCHITECTURE

We wrote the embedded microcontroller firmware using the Arduino IDE in C++. We had to be very careful with memory and timing:

• **Non-Blocking FreeRTOS Tasks:** We never used Arduino's `delay()` function. When `delay()` runs, the processor stops completely, which freezes the screen and misses button presses. Instead, we used FreeRTOS timers and non-blocking `millis()` loops.

• **Solving the Brownout Crash:** When we first tested the ESP32-CAM, every time it tried to take a photo while transmitting Wi-Fi, the board crashed and rebooted! We used an oscilloscope in our school physics lab and discovered that the 3.3V rail dipped down to 2.4V for 15 milliseconds. We soldered a 470 uF electrolytic capacitor directly across the 5V and GND pins of the camera. That solved the brownout completely.

• **Float16 Quantization:** We trained MobileNetV3-Small on our curated leaf dataset. The standard model was over 8.5 MB, which was too sluggish for cheap smartphones. Using TensorFlow Lite's Float16 post-training quantization, we shrunk the model down to **2.03 MB** (SHA-256 hash: be71ededc...). On our reference interpreter, it executes in just **4.94 ms**, and takes around 12–15 ms on a budget Samsung Galaxy A03.

19. HOW A FARMER USES THE WAND (4-PAGE OLED CAROUSEL)

To make the device completely easy for an elderly or non-English speaking farmer, we programmed the SSD1306 display to present results across four clean screens that flip sequentially when the farmer clicks the 'NEXT' button:

• **Page 1 (Primary Diagnosis):** Shows the crop name, predicted health condition in large clear letters, and the confidence score (e.g., POTATO: LATE BLIGHT [89%]). If the model is unsure, it displays UNCERTAIN / RESCAN.

• **Page 2 (Canopy Microclimate):** Displays the real-time temperature and relative humidity measured by the DHT11 inside the foliage (e.g., TEMP: 21.4°C | RH: 88%) along with an immediate risk flag (HIGH BLIGHT RISK).

• **Page 3 (Agronomic Prescription):** Shows an immediate, actionable organic or chemical advice in plain Hindi/English

words (e.g., ACTION: SPRAY TRICHODERMA OR COPPER OXYCHLORIDE (2g/L); ISOLATE SICK LEAVES).
• **Page 4 (System Telemetry):** Shows battery voltage, Wi-Fi link strength (RSSI), and total images stored on the SD card.

20. METHODOLOGY (ENGINEERING DESIGN CYCLE)

We followed the classical Engineering Design Cycle throughout the five months of this project:
1. *Problem Definition:* Investigated multi-state crop destruction tracked from agricultural news and farmer channels; surveyed local vegetable growers.
2. *Breadboard Prototyping:* Wired initial circuits on breadboards; encountered power dropouts and camera focus failures.
3. *Optical Redesign:* Built the cylindrical shroud to solve camera focus and remove background clutter.
4. *Cryptographic Data Curation:* Discovered duplicate images across public datasets; wrote Python scripts to clean and isolate holdouts.
5. *Field Testing & Verification:* Tested 60 fresh botanical leaves and 93 challenging outdoor photos; analyzed every error honestly.

21. DATASET CLEANING AND CRYPTOGRAPHIC AUDIT

A major breakthrough in our project came when we audited our training dataset. In many student AI projects, students blindly download thousands of images and claim 99% accuracy. But when we checked the images using SHA-256 cryptographic hashes and perceptual dHash (difference hashing), we discovered that standard datasets like PlantVillage contain identical duplicate images in both training and test folders! If a model sees the exact same image during testing, it is just memorizing, not learning.

We wrote an automated audit script that strictly reconciled our 7,710 dual-domain records into **6,965 unique cryptographic hashes**. We completely purged all near-duplicate images within a dHash Hamming distance of 2 from our 60 benchmark leaves. This gave us absolute certainty: our test benchmarks were 100% unseen by the neural network during training.

22. TRAINING AND TESTING METHODOLOGY

We trained MobileNetV3-Small across six balanced plant health classes: (1) Tomato Early Blight, (2) Tomato Late Blight, (3) Tomato Healthy, (4) Potato Early Blight, (5) Potato Late Blight, and (6) Potato Healthy.

We applied realistic data augmentations: random rotations (±15°), slight zoom changes (0.9x to 1.1x), and subtle brightness variations to mimic the diffuse LED lighting inside the shroud. We used an Adam optimizer with an initial learning rate of 0.001 and categorical cross-entropy loss. We set an automated checkpoint to save only the model weights that minimized validation loss. The final trained model was exported to TensorFlow Lite format and locked with SHA-256 integrity verification.

23. PHYSICAL FIELD EXPERIMENTS AND OBSERVATIONS

We conducted physical tests under four real-world conditions:
• **Working Distance Test:** Without the shroud, when holding the camera by hand at 4 cm, images were blurred macro smudges; at 15 cm, background mud and dry straw filled 60% of the frame. With our 8 cm shroud, every single photo framed exactly 85–90% leaf surface in sharp focus.
• **Lighting Comparison:** In direct midday sunlight, unshaded leaves produced bright specular white spots that the neural network confused with powdery mildew. Inside the shroud, the diffuse LED provided uniform 350-lux illumination

with zero glare.
• **Battery Life:** Powered by a standard 5,000 mAh portable USB power bank, the entire dual-wand system ran continuously for **6 hours and 42 minutes**, allowing more than 400 complete leaf scans on a single charge.

24. RESULTS AND THEIR SCIENTIFIC INTERPRETATION

We evaluated our system against two completely separate, independent test sets without any data leakage:

A. Clean Botanical Benchmark (60 Fresh Leaves):
Across 60 carefully curated leaves (10 per class), KRISHI-DRISHTI achieved **88.33% strict six-class accuracy (53 out of 60 correct)**. When our Shannon entropy uncertainty filter was turned on (threshold H > 1.45 nats), it flagged 3 highly ambiguous leaves as 'UNCERTAIN', raising the accepted diagnostic accuracy to **92.98% (53 correct out of 57 accepted)**! Also, in 54 out of 60 cases (**90.0%**), the model correctly identified the pathogen family (e.g. recognizing Late Blight oomycete pathology even if it momentarily confused potato with tomato).

B. Open-Air Complex Benchmark (93 PlantDoc Field Images):
To test how our system handles messy real-world photography, we tested it against 93 genuine outdoor leaf images from the PlantDoc dataset. These images contain dirt, insect bites, and background weeds. The raw unguided model scored only **33.33% (31 out of 93 correct)**. This proved that laboratory AI models struggle when thrown into unconstrained outdoor dirt! However, when our dual-stage Shannon entropy filter was engaged, it safely rejected 37 confusing images as UNCERTAIN. On the remaining 56 accepted images, accuracy jumped to **55.36% (31 out of 56 correct)**. This proved our hypothesis: uncertainty gating prevents harmful false advice.

Evaluation Benchmark Dataset	Total Images	Raw Correct	Raw Accuracy	Gated Uncertain	Accepted Accuracy
Clean Botanical Benchmark (Holdout)	60	53	88.33%	3 leaves	**92.98%** (53/57)
Pathogen-Family Match (Biological)	60	54	90.00%	—	**90.00%** (54/60)
PlantDoc Open-Air Dataset (Messy)	93	31	33.33%	37 leaves	**55.36%** (31/56)

25. COMPARISON TABLES AND HAND-DRAWABLE VISUALS

Hand-Drawable Pie Chart: Real-World Distribution of Field Diagnostic Scans:
When small farmers screen their crops, every leaf does not fit into a neat textbook picture. Based on our 150 total field trials, scans naturally divide into three distinct categories. We calculated the exact protractor angles so any student can draw this pie chart neatly:
• *High Confidence Clear Diagnosis:* 50.0% of scans --> **Protractor Angle = 180°**
• *Gated Uncertain / Rescan Needed:* 26.9% of scans --> **Protractor Angle = 97°**
• *Pathogen Family Match (Guidance):* 23.1% of scans --> **Protractor Angle = 83°**
(Verification: 180° + 97° + 83° = Exactly 360.0° Complete Circle)

26. REAL-WORLD LIMITATIONS AND HONEST FAILURE MODES

As young student scientists, we believe that true science requires complete honesty about mistakes and failures. We do not claim our project is a magic, flawless device. During our physical testing, we discovered two serious failure modes:

1. The Out-of-Distribution (OOD) Alien Leaf Trap:
We wondered: what happens if a farmer accidentally puts a leaf from a completely different plant into the shroud? We

plucked a healthy, dried leaf from a nearby tea plant (*Camellia sinensis*) and placed it inside the tube. To our astonishment, the neural network did not say 'unknown plant'. Instead, it confidently declared **'Potato Late Blight' with 87.0% confidence!** This was a huge eye-opener for us. It proved that a neural network trained only on tomatoes and potatoes will force every visual pattern into one of its six trained buckets. Our optical shroud enforces physical distance and lighting geometry, but it does NOT grant biological intelligence. A farmer must ensure they are only scanning potato or tomato foliage!

2. Drop-Shadow False Alarms:
When we tested leaves in the dark using an angled flashlight, the leaf vein cast a dark drop-shadow onto the green blade. The neural network saw this dark curve and misclassified a perfectly healthy potato leaf as Early Blight with 97.9% confidence! To fix this, we added an acrylic diffusion disc over the LED to soften harsh point shadows.

27. SAFETY AND ENVIRONMENTAL CONSIDERATIONS

• **Electrical & Battery Safety:** The system runs on 5V DC low-voltage power supplied by an insulated USB power bank. There is zero risk of high-voltage electric shock, even when operating in humid, damp field environments.
• **Non-Destructive Testing:** The optical shroud is simply held against the leaf blade while the plant is growing. The farmer does not need to tear off leaves or damage the crop.
• **Soil and Pollinator Protection:** By helping farmers recognize that early spots are often harmless or manageable with bio-fungicides (like *Trichoderma viride* or diluted neem oil), our system prevents unnecessary chemical drenching of soil and protects wild honeybees and soil earthworms.

28. SUSTAINABILITY AND EXPECTED USEFULNESS TO SMALL FARMERS

KRISHI-DRISHTI is built with sustainability at its heart:
1. **Zero Recurring Expenses:** Once built, the wand does not require monthly subscriptions, cloud server fees, or data recharges. It can be shared by an entire village cooperative or farmer self-help group (SHG).
2. **Huge Household Savings:** A single wasted fungicide spray costs a small farmer Rs 1,200 to Rs 2,500 per acre. Preventing even two unnecessary sprays pays for the entire cost of our wand in less than one season.
3. **Local Repairability:** Every component used (ESP32, OLED, DHT11, wires) is widely available in district town electronic markets. If a wire comes loose, any local mobile repair technician or high school student can re-solder it in five minutes.

29. CONCLUSION

Through this Class 9 research investigation, we proved that advanced Edge-AI and IoT technology does not have to be expensive or dependent on foreign cloud servers. By pairing low-cost microcontrollers with a simple physical optical shroud, lightweight quantized neural networks, and canopy microclimate sensing, we built a working, field-ready screening wand for under Rs 1,500.

Our device achieved 88.33% strict accuracy and 92.98% accepted accuracy on clean botanical samples, and successfully filtered out 37 confusing images on outdoor tests using Shannon entropy gating. While it cannot replace laboratory microscopes or common sense, KRISHI-DRISHTI gives smallholder farmers an honest, affordable first line of defense to protect their harvests sustainably.

30. FUTURE IMPROVEMENTS (OUR NEXT MILESTONE)

As we continue developing this system, we plan to implement three specific improvements:
1. **On-Chip Neural Inference:** Port an ultra-compact int8 neural network directly onto the ESP32-S3 chip using the

ESP-NN library, eliminating the need for an external smartphone completely.
2. **Expand Crop Palette:** Add training classes for wheat rust and mustard blight, which are also major winter crops in northern India.
3. **Multi-Spectral Near-IR Sensing:** Add an inexpensive 850 nm infrared LED to detect internal leaf cellular breakdown before visible black spots even appear on the outside surface.

31. ACKNOWLEDGEMENTS

We express our heartfelt gratitude to our Science Guide Teacher for teaching us plant pathology concepts, guiding our circuit wiring, and encouraging us when our first prototypes failed. We sincerely thank our School Principal for granting us access to the physics lab and 3D printer. We are deeply grateful to the local vegetable farmers who patiently allowed us into their fields to test our camera wand and share their farming struggles. Finally, we thank our parents for their constant support and encouragement.

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